104 Changes of State

1.4.1 · 1.4.2 · 1.4.3 Changes of state and latent heat

1.4.1 Latent heat absorbed in melting and vaporisation

Melting and vaporisation require energy to change particle arrangements or separation. A pure substance changing state at its melting or boiling point under fixed pressure remains at constant temperature while absorbing latent heat. No temperature rise does not mean no energy transfer. Further heating raises the temperature once the change of state is complete.

1.4.2 Latent heat released in freezing and condensation

Freezing and condensation release latent heat to the surroundings as particle arrangements or separations change and internal energy falls. A pure substance can change state at constant temperature under fixed pressure. For the same substance under the same conditions, reverse changes of state absorb and release equal amounts of latent heat per kilogram.

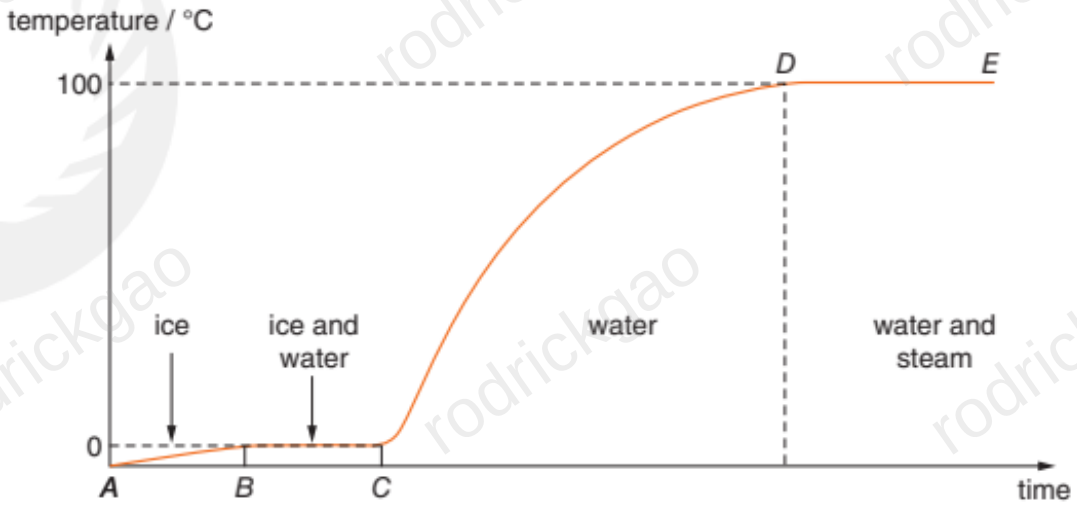
1.4.3 Specific latent heat

Specific latent heat is the energy absorbed or released per kilogram undergoing a specified change of state. Multiply it by the mass actually changing state to obtain the energy transferred. Distinguish fusion from vaporisation. If a sample first warms and then changes state, calculate the temperature-change energy and latent heat separately.

$$l = \frac{Q}{m}$$

$$Q = ml$$

l: specific latent heat (J kg⁻¹); Q: energy absorbed or released (J); m: mass changing state (kg). Pure substance at its transition temperature and fixed pressure.



Horizontal sections of the heating curve mark changes of state: energy is absorbed without a temperature rise.

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1.4.4 Internal energy

Internal energy is the sum of the kinetic energies of random microscopic particle motion and the potential energies of particle interactions. It excludes the kinetic energy of the object's bulk motion and its overall gravitational potential energy. Heating usually increases mean particle kinetic energy, whereas a change of state can change internal energy without changing temperature.

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